# Firms That Look Alike: Characteristic Graphs and the Cross-Section of Stock Returns

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**Abstract**

Graph neural networks (GNNs) forecast stock returns by passing information between economically or statistically linked firms. We ask whether links inferred from 64 firm characteristics can improve return forecasts relative to a neural network that uses the same characteristics without a graph. In U.S. stocks from 2010 to 2024, a simple neighbor graph is not consistently superior to a basic neural network, and a learned neighbor graph results in a positive annual-return difference for one specification, but not consistently across a range of specifications. This is mirrored in the raw comovement between characteristic neighbours, with the characteristics used to define them accounting for much of the information in the raw data, leaving less information to be leveraged by graph-based message passing. This may be because characteristic-based neighbours are more likely to serve as a reference group than as a source of extra information: in the reported specification, the annualised volatility of the plain network drops from 23% to 9%, while the Sharpe ratio increases from 0.69 to 1.03 when ranking the forecasts within 20 characteristic clusters. A strategy using the previous month's return of a firm compared with its 20 closest characteristic-based neighbours shows peer-relative return reversal with a gross Sharpe ratio of 1.38, and the signal is focused in the short leg and small firms.

**Keywords:** graph neural networks, firm similarity, return predictability, short-term reversal, look-ahead bias. **JEL:** C45, C55, G11, G12, G14.

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## 1 Introduction

Return predictability across firms with economic links is well established in a large literature, and is consistent with different firms having different rates of information diffusion. Such predictability has been documented among customers and suppliers (Cohen and Frazzini, 2008), firms in related industries (Menzly and Ozbas, 2010), technologically related firms (Lee et al., 2019), and firms covered by the same analysts (Ali and Hirshleifer, 2020). This can be extended to graph neural networks (GNNs) that combine information from connected companies to generate return forecasts (Scarselli et al., 2009; Kipf and Welling, 2017; Wu et al., 2022). Capponi et al. (2026) use graph neural networks to model supplier–customer relationships and find that the improvements are significant, when compared with a network that does not include these links.

Observed economic links are not consistently available for every firm and period. Thus, we study graphs built from firm characteristics that can serve to define the neighbours of a firm across time and firms. Two companies that are similar in size, valuation, profitability, investment, momentum and volatility can have the same risk factors or investor base. The crux question is whether combining data from such companies is benefiting the prediction compared to a neural network monitoring the same features. If so, a GNN could learn the relevant links from the 64 characteristics of Freyberger et al. (2020) alone.

Figure 1 shows what such a graph contains. Linking each firm to its ten most similar firms recovers much of the industry structure without using industry codes: 41% of links join firms in the same two-digit industry, against 15% if links were random (panel a). These patterns suggest that characteristic similarity captures some aspects of economic proximity. Whether it adds predictive *information* is the central question of this paper.

We address this question in four steps: comparing fixed and learned characteristic graphs with a plain neural-network benchmark; testing whether within-cluster ranking will enhance portfolio construction; testing the geometry of learned representation and its predictive performance; testing whether there is additional return comovement among characteristic neighbours.

We also know that the previous month's return of a firm compared to its 20 most similar

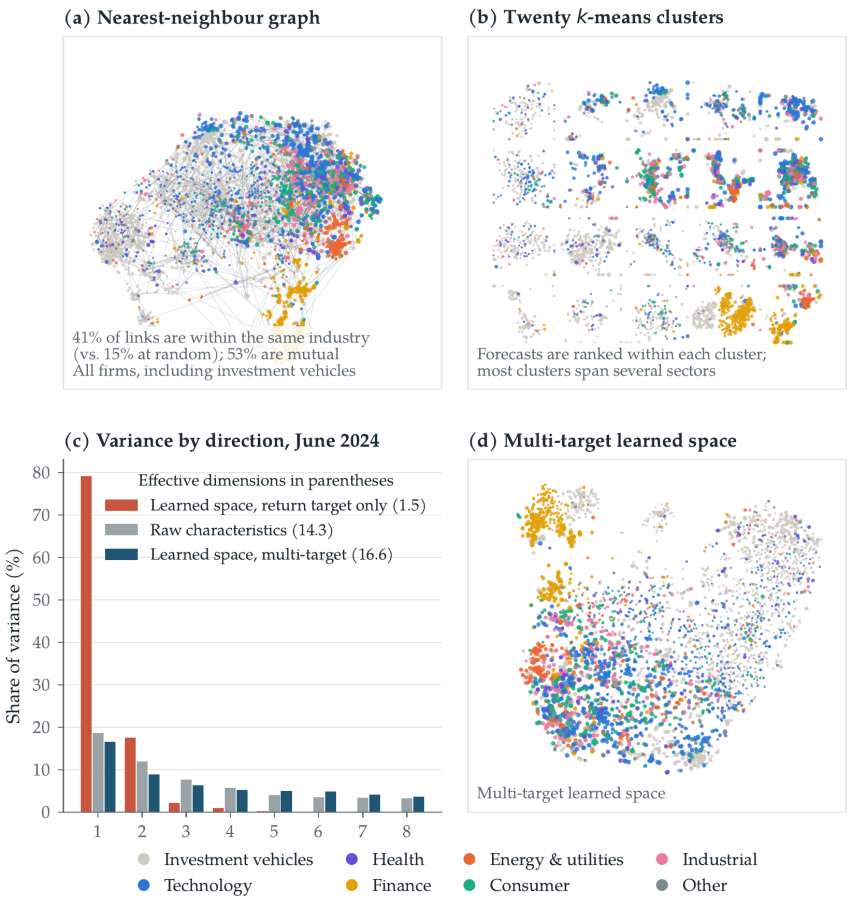



**Figure 1. The cross-section of U.S. stocks as a similarity map, June 2024 (3,876 firms).** (a) Two-dimensional $t$-SNE projection of the 64 characteristic ranks; lines join a random sample of firms to their three nearest neighbours by cosine similarity; marker area increases with market capitalization; colour denotes sector. (b) The same firms grouped into the 20 monthly $k$-means clusters used in Section 5.2. (c) Share of variance in the first eight principal directions of three similarity spaces; effective dimensions are the participation ratio $(\sum_k \lambda_k)^2 / \sum_k \lambda_k^2$. (d) $t$-SNE projection of the multi-target learned space of Section 5.3.

neighbours on the basis of the firm's characteristics predicts the next month's return with a gross Sharpe ratio of 1.38 ($t = 4.1$). The signal is concentrated in the short leg and among smaller firms; monthly data cannot establish whether the result reflects short-lived reversal or bid–ask bounce. It should therefore be interpreted as evidence of peer-relative short-term reversal, not as an established estimate of implementable trading profits (Jegadeesh, 1990; Lehmann, 1990; Da et al., 2014; Hameed and Mian, 2015).

The analysis is informed by three data audits: the timing of return on capital (ROC) is corrected, a wider union panel is used to account for survivorship concerns with the previous panel, and the evolution of the investment vehicles and unclassified firms is explored. These issues and the data choices are described in section 3.

## 2 Related literature

**Return prediction and economic links.** Machine-learning techniques can pick up on nonlinearities and interactions between firms' features that linear models might not. Gu et al. (2020) show that trees and neural networks perform particularly well in this setting, while Kozak et al. (2020) and Chen et al. (2024) address high-dimensional prediction through regularization and economically structured deep learning. Our question is whether aggregating information across firms adds predictive value beyond a model that already uses their characteristics.

A related literature shows that returns predictably propagate across economically connected firms. Cohen and Frazzini (2008) use customer–supplier links, Menzly and Ozbas (2010) study cross-predictability across related industries, and Lee et al. (2019) identify return predictability among technologically linked firms. Ali and Hirshleifer (2020) study firms sharing analyst coverage, while Hoberg and Phillips (2016) construct product-market peers from firms' 10-K descriptions. These approaches use observed economic relationships or textual similarity. Capponi et al. (2026) apply graph neural networks to supply-chain relationships. Instead, we build static and handcrafted graphs based on firm attributes and verify if message passing is able to outperform a matched neural-network benchmark without the use of graph aggregation.

This distinction also matters for interpreting return comovement. Daniel and Titman (1997) emphasise the role of characteristics in explaining average returns, whereas Kelly et al. (2019) show how characteristics can proxy for exposures to latent risk factors. Because these characteristics cause firms to have similar exposures, firms with similar characteristics may comove. We test if the characteristics of the neighbours, except for the characteristics used to define the neighbours, show additional comovement.

**Graph neural networks.** Graph neural networks learn firm- or node-level representations by repeatedly aggregating information from neighbouring nodes (Scarselli et al., 2009; Wu et al., 2021, 2022). Graph convolutions (Kipf and Welling, 2017), inductive neighbourhood aggregation (Hamilton et al., 2017), attention over neighbours (Veličković et al., 2018) and the general message-passing framework (Gilmer et al., 2017) underlie the architectures we use, and node-level prediction with these models is reviewed by Tang and Liao (2022). The expressive

power of message passing is bounded by the Weisfeiler–Lehman graph-isomorphism test (Xu et al., 2019; Li and Leskovec, 2022), and deep stacks tend to over-smooth node representations (Li et al., 2018; Oono and Suzuki, 2020; Chen et al., 2020a). When the graph is not observed, its structure can be learned jointly with the predictor (Franceschi et al., 2019; Chen et al., 2020b; Chen and Wu, 2022), for example by recomputing nearest neighbours in a learned feature space (Wang et al., 2019); graphs that change over time are reviewed by Kazemi (2022) and auxiliary self-supervised objectives by Wang et al. (2022). Our learned graph follows the structure-learning approach, with neighbours recomputed each month from the firms' learned representations.

**Learned representations and portfolio construction.** Interpretable machine-learning research shows that flexible models can retain economically informative structure (Bell et al., 2026c; Liu et al., 2022). Bell et al. (2026b) and Bell et al. (2026a) apply interpretable machine learning to corporate-bond returns and equity portfolio optimisation, respectively, distinguishing return prediction from portfolio construction. Forecasts are assumed to be fixed and we compare the ranking for all the firms with ranking within characteristic clusters; the portfolio improvement is because of the ranking rule and NOT because of improved forecasts Our exercise goes beyond AlphaGlass training the model directly on a portfolio objective; rather, it changes portfolio construction after prediction.

The meaning of an economic learned graph also relies on the definition of similarity that is represented.Text-based measures, such as those in Hoberg and Phillips (2016) and Kakhbod et al. (2024), capture product-market or technological relationships rather than accounting-characteristic similarity. Related text- and language-model-based methods have been used to study Federal Reserve policymaking (Kakhbod et al., 2026) and citation gaps for patents with female inventors (Hochberg et al., 2026), while experimental evidence examines how AI interacts with investors' perception biases (Fedyk et al., 2024). We assess learned characteristic spaces with the effective dimensionality and neighbourhood stability, but imagine them simply as a diagnostic assessment, not evidence that the learned neighbours are meaningful economically.

**Information timing.** The information for any predictive representation used must be available at the forecast time. Kakhbod and Li (2025) address look-ahead and look-back bias in language-

model research by restricting pretraining to timestamped historical text. This is the same for tabular predictors. In our data, return on capital (ROC) was correlated with the return month, thus generating a timing effect which enhanced the apparent predictive value. We apply ROC before using it in the main analysis (Section 3); this correction is necessary to assess the graph models with the proper information set.

## 3 Data and audits

We use a firm-month panel of U.S. common stocks containing the 64 characteristics of Freyberger et al. (2020), each represented as a cross-sectional rank in $[0, 1]$, together with the two-digit SIC code, monthly return and risk-free rate. The panel contains 1,065,779 firm-month observations for 10,426 firms from January 2003 to December 2024. The model input contains 64 characteristics. `Spread` is constant across firms within each month and therefore provides no cross-sectional discrimination. A row dated $t$ contains characteristics intended to be known at the start of month $t$ and the return realized during month $t$. The one-month-lagged return characteristic, $r_{2,1}$, correlates at least 0.985 with the previous month's return rank in each of the 263 months in which the lag can be tested. January 2003 is the first month in the panel and cannot be validated against an earlier observation; in that month, $r_{2,1}$ equals the rank of the current month's return (Spearman correlation 1.00). We treat this value as missing in the corrected comovement analysis. The original model training samples included the January 2003 observation and were not re-estimated, so the effect of this anomaly on fitted parameters cannot be assessed. The prediction target is the month-$t$ excess return, winsorized at the 1st and 99th percentiles and demeaned within month; portfolios use unwinsorized returns. Three audits inform the data choices: survivorship in an earlier panel, the timing of ROC, and the growing share of investment vehicles and unclassified firms.

**Survivorship.** An earlier panel from the same source (1977–2024) retains 83.0% of its 5,540 firms in its final month, and the 942 firms that exit earn $-8.9\%$ on average in their last month. In the panel used here, 36.0% of firms survive to the final month and exiting firms earn $-0.7\%$. We use the broader union panel for the main analysis, but have not verified whether it includes

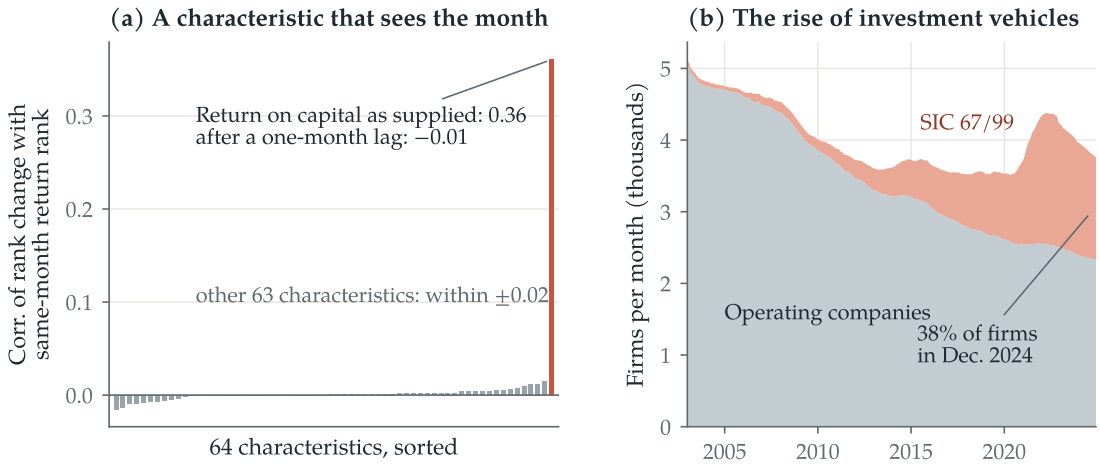



**Figure 2. Data audits.** (a) For each characteristic, the mean monthly correlation between the change in its cross-sectional rank and the rank of the same month's return, January 2003–December 2024, sorted. A characteristic known at the start of the month should lie near zero. (b) Number of firms per month, split into operating companies and firms in SIC 67 (holding and other investment offices) or 99 (unclassifiable).

delisting returns (Shumway, 1997).

**A characteristic that reflects the current month.** The change in a characteristic's rank should be uncorrelated with the return realized during the same month if the characteristic is known at the start of that month. For 63 characteristics, the average correlation lies within ±0.016; for return on capital (ROC), it is 0.361 (Figure 2a). Lagging ROC by one month reduces this correlation to approximately zero. In a walk-forward ridge regression, the long–short Sharpe ratio falls from 7.59 with ROC as supplied to 0.75 after the correction, while the information coefficient falls from 0.177 to 0.066. We therefore lag ROC throughout; 1.01% of rows lack a previous value and receive the median rank.

**Investment vehicles.** Firms in SIC 67 and 99 rise from 3.7% of the cross-section in January 2010 to 41.1% in January 2022 and 38.2% in December 2024 (Figure 2b). Their returns are easier to rank: in 2021, the ridge information coefficient is 0.21 among these firms and 0.11 among operating firms. We therefore report results both including and excluding these firms.

## 4 Models and evaluation

All models share the same base encoder; graph models add information aggregated from other firms (Figure 3).

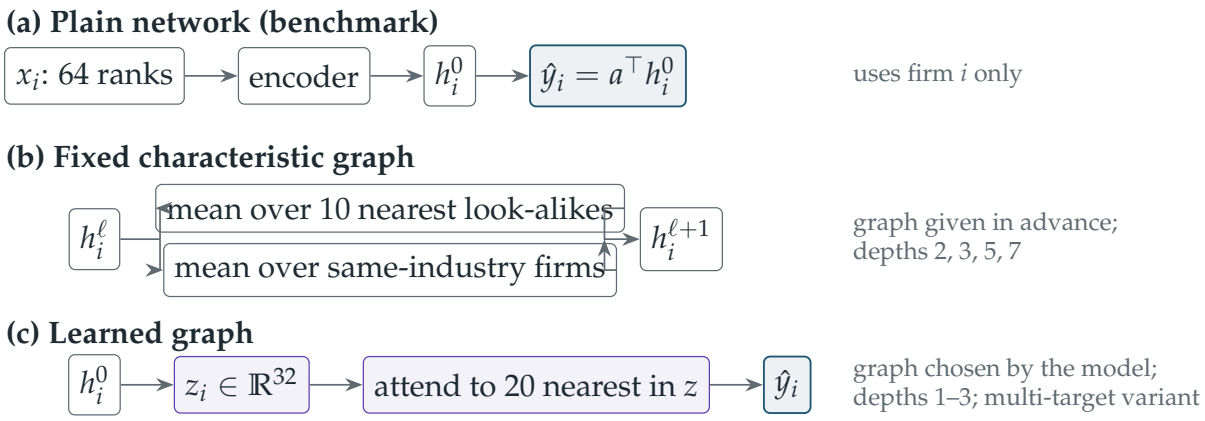



**Figure 3. Three ways to use other firms.** All models share the encoder in panel (a); a graph model at depth zero is exactly the plain network.

**Plain network.** With $x_{i,t} \in [-\frac{1}{2}, \frac{1}{2}]^{64}$ the centred characteristic ranks (missing values set to zero), the encoder is $h^0_{i,t} = \phi\big(W_2\,\delta(\phi(W_1 x_{i,t} + b_1)) + b_2\big) \in \mathbb{R}^{64}$, where $\phi$ is the ReLU and $\delta$ dropout, and the plain network predicts $\hat{y}_{i,t} = a^\top h^0_{i,t} + a_0$.

**Fixed graph.** Firm $i$'s neighbours $\mathcal{N}_t(i)$ are its ten most similar firms by cosine similarity of $x_{i,t}$, and $\mathcal{I}_t(i)$ is its two-digit industry. A GraphSAGE-style layer (Hamilton et al., 2017) updates

$$h_i^{\ell+1} = \mathrm{LN}\Big(h_i^\ell + \delta\big(\phi(W_s^\ell h_i^\ell + W_n^\ell \bar{h}^\ell_{\mathcal{N}(i)} + W_d^\ell \bar{h}^\ell_{\mathcal{I}(i)})\big)\Big), \tag{1}$$

where $\bar{h}_{\mathcal{S}}$ is the mean over set $\mathcal{S}$ and LN is layer normalization. Depth $L$ is the number of hops information travels.

**Learned graph.** Each firm is placed at $z_{i,t} = \tilde{q}_{i,t}/\|\tilde{q}_{i,t}\|$, where $\tilde{q}_{i,t} = Qh^0_{i,t} - N_t^{-1}\sum_j Qh^0_{j,t}$, and attends to the 20 firms with the highest similarity $s_{ij} = z_i^\top z_j$ with weights $\alpha_{ij} \propto \exp(\tau s_{ij})$ and learned temperature $\tau$ (Veličković et al., 2018; Chen et al., 2020b). A layer forms $m_i = \sum_j \alpha_{ij}\,\phi(U[h_j;\, h_j - h_i] + c)$ and updates $h_i \leftarrow \mathrm{LN}(h_i + \delta(\phi(Sh_i) + m_i))$. The forecast is $\hat{y}_i = a^\top h_i^0 + g^\top h_i^L$ with $g$ initialized at zero, so that training starts at the plain network. The multi-target variant, at depth 2, adds two objectives to the loss: reconstruction of the input characteristics from $h_i^0$, and agreement between learned similarity and realized firm comovement over the preceding 12 months, using returns dated before the test block (Wang et al., 2022). A separate query–key version collapsed onto a few hub firms; because this version did not produce a useful distribution of neighbours, we do not use it as evidence that the learned graph captures

meaningful similarity.

**Training and evaluation.** Models are refit every two years using an expanding training window. For each test block, the immediately preceding 36 months are reserved for early stopping and excluded from gradient-based fitting; no test-period observations enter training or validation. We use Adam (learning rate $10^{-3}$, weight decay $10^{-4}$), dropout 0.3, hidden width 64, and one month's cross-section per gradient step, and we average forecasts over independently initialized runs. Each month we sort firms into deciles on the forecast and hold the equal-weighted decile 10 minus decile 1. "Within $k$ clusters" ranks the forecast inside the firm's $k$-means cluster of that month's characteristics, so that each cluster contributes its own long and short positions. We report annualized means, volatilities and Sharpe ratios, $t$-statistics with six lags (Newey and West, 1987), turnover, Sharpe ratios net of an assumed transaction cost of 20 basis points per unit of one-way turnover, and Sharpe ratios after excluding firms in the bottom size quintile. The transaction-cost adjustment is illustrative rather than empirically estimated because order-book and quote data were unavailable. The test period is January 2010 to December 2024 (180 months). All statistics were recomputed from saved forecasts or by rerunning saved model weights.

## 5 Look-alikes in forecasting models

### 5.1 A fixed similarity graph does not improve forecasts

Table 1 compares the fixed-graph GNN with the plain network. A single-run depth sweep (Panel A) suggests that depth 3 is best, with a Sharpe ratio of 0.90 against 0.74. This comparison is exploratory: the depth-3, depth-5 and depth-7 results in Panel A are each based on one run, so the apparent depth-3 advantage should not be interpreted as robust. With three runs per model and ROC lagged (Panel B), the plain network earns a Sharpe ratio of 0.81 and the depth-3 GNN 0.72, a difference of $-1.8\%$ a year ($t = -1.04$). At depth 7 the GNN performs worse (0.35), consistent with over-smoothing (Li et al., 2018), in which repeated aggregation makes firm representations less distinct (Figure 1a). When the Panel B models are evaluated on operating firms only, the depth-3 GNN also underperforms the plain network: its Sharpe ratio within 20 clusters is 0.70 against 1.14, a difference of $-3.0\%$ a year ($t = -2.28$). Investment vehicles

**Table 1.** Fixed-graph GNN and plain network: long–short Sharpe ratios, 2010–2024

| Model | Ranked across all firms Sharpe | ($t$) | Within 10 clusters | Within 20 clusters | Within SIC2 |
|---|---|---|---|---|---|
| *Panel A. Depth sweep, ROC dropped, vehicles included* | | | | | |
| Ridge regression | 0.76 | (2.43) | 1.32 | 1.15 | 0.83 |
| Plain network (depth 0) | 0.74 | (2.37) | 1.21 | 1.45 | 0.92 |
| Fixed graph, depth 2 | 0.61 | (1.95) | 1.05 | 1.09 | 0.77 |
| Fixed graph, depth 3 | 0.90 | (2.78) | 1.44 | 1.39 | 1.08 |
| Fixed graph, depth 5 | 0.63 | (1.84) | 0.99 | 0.93 | 0.83 |
| Fixed graph, depth 7 | 0.35 | (1.09) | 0.32 | 0.22 | 0.45 |
| *Panel B. ROC lagged, vehicles included, three runs per model* | | | | | |
| Ridge regression | 0.75 | (2.39) | 1.12 | 1.18 | 0.82 |
| Plain network | **0.81** | (2.70) | 1.29 | **1.42** | 0.96 |
| Fixed graph, depth 3 | 0.72 | (2.26) | 0.97 | 1.10 | 0.74 |
| *Panel C. Panel B models, evaluated on operating firms only* | | | | | |
| Plain network | **0.83** | (2.74) | 1.13 | **1.14** | – |
| Fixed graph, depth 3 | 0.59 | (1.99) | 0.55 | 0.70 | – |

Equal-weighted decile long–short portfolios before costs; Newey–West $t$-statistics (6 lags). Panel A: plain network and depth 2 averaged over three runs, depths 3, 5 and 7 one run; values from the summary saved by the original runs. Panels B and C recomputed from saved forecasts; in Panel C clusters are refit on operating firms.

therefore account for part of the fixed graph's apparent performance, as they occupy dense, homogeneous regions of the graph (Figure 1a). Overall, the matched results do not show a robust forecasting benefit from the fixed graph (Figure 1a).

### 5.2 Ranking within clusters

The largest improvement comes from changing portfolio construction, not the forecasts themselves. Figure 4a compares the plain network's forecasts ranked across all firms or within 10 or 20 monthly characteristic clusters. Ranked across all firms, the portfolio has large common exposures: its long and short legs tilt toward different characteristic profiles, a tilt that was profitable for much of the decade but reversed in 2020–2021, producing a maximum drawdown of −72%. Ranked within 20 clusters, firms are compared with others of similar characteristics. Annualized volatility falls from 23.3% to 9.2%, maximum drawdown from −72% to −23%, and the Sharpe ratio rises from 0.69 to 1.03. At a common 10% volatility, the same forecasts earn 10.3% a year within 20 clusters, compared with 6.9% across all firms.

The improvement is present in most, but not all, specifications (Figure 4a and Table 1). Ranking within clusters removes some shared exposure while retaining the model's ability to

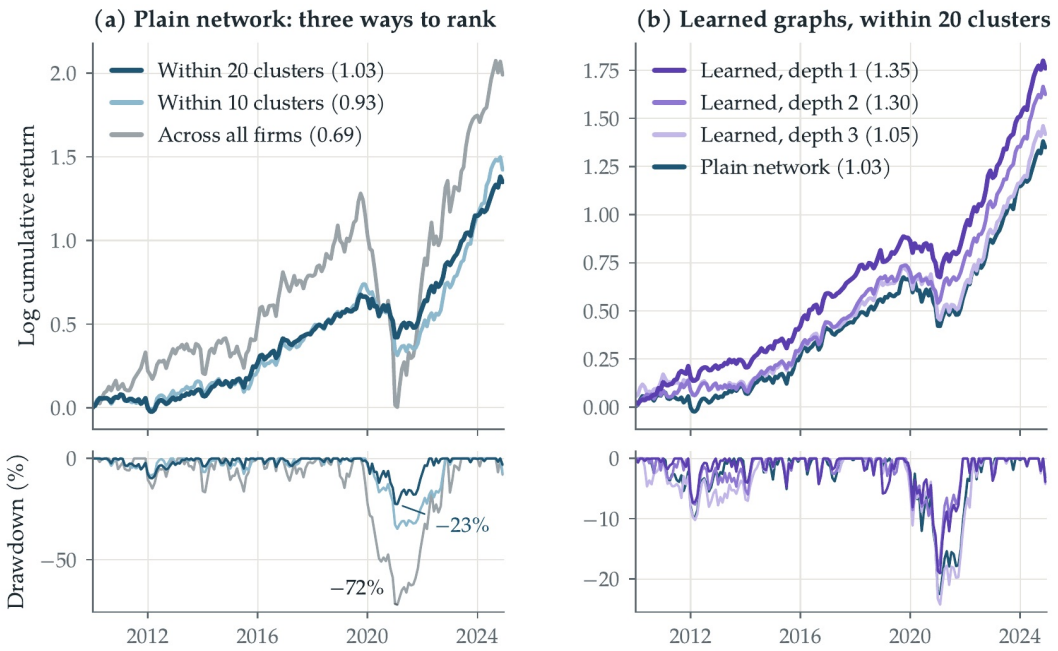



**Figure 4. Ranking within clusters and learned graphs.** Decile long–short portfolios, investment vehicles included, January 2010–December 2024. Top: log cumulative return; bottom: drawdown from the previous peak. Sharpe ratios in parentheses. (a) The plain network's forecasts, ranked across all firms or within monthly characteristic clusters; the forecasts are identical in all three portfolios. (b) Learned-graph models and the plain network, all ranked within 20 clusters.

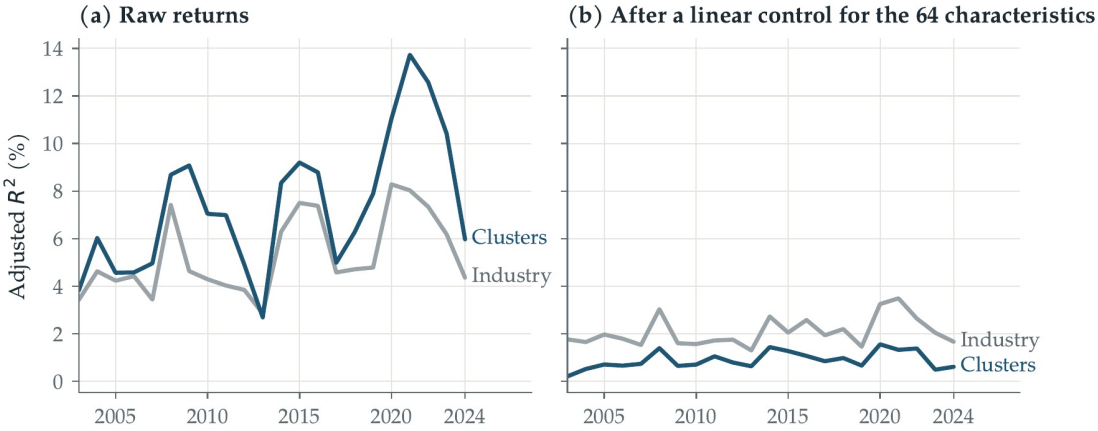



**Figure 5. Every strategy on one scale.** Annualized Sharpe ratios of decile long–short portfolios before costs, investment vehicles included, January 2010–December 2024. For each model the three markers correspond to the three ranking rules; the number is the Sharpe ratio within 20 clusters. The two strategies below the dotted line estimate no model.

discriminate among firms with similar characteristics. This is consistent with Bell et al. (2026a), who emphasize the role of portfolio construction in the design of predictive strategies. The number of clusters was chosen after inspecting an earlier panel and is not validated out of sample, so the size of the improvement should be interpreted with this selection caveat.

**Table 2.** Learned-graph models, 2010–2024

| | Return (%/yr) | Vol. (%/yr) | Sharpe | ($t$) | Max DD (%) | Turn. (%/mo) | Sharpe net 20 bps | Sharpe ex small | vs. plain: %/yr ($t$) |
|---|---|---|---|---|---|---|---|---|---|
| *Vehicles included, ranked across all firms* | | | | | | | | | |
| Plain network | 16.1 | 23.3 | 0.69 | (2.18) | −72 | 37 | 0.61 | 0.43 | – |
| Learned, depth 1 | 17.4 | 23.4 | 0.75 | (2.47) | −65 | 38 | 0.67 | 0.43 | 1.3 (0.96) |
| Learned, depth 2 | 17.8 | 22.5 | 0.79 | (2.58) | −66 | 39 | 0.71 | 0.52 | 1.6 (1.09) |
| Learned, depth 3 | 14.5 | 24.3 | 0.60 | (1.90) | −74 | 40 | 0.52 | 0.37 | −1.6 (−1.00) |
| Multi-target, depth 2 | 16.2 | 27.6 | 0.59 | (2.09) | −73 | 32 | 0.53 | 0.44 | 0.1 (0.04) |
| *Vehicles included, ranked within 20 clusters* | | | | | | | | | |
| Plain network | 9.4 | 9.2 | 1.03 | (3.77) | −23 | 63 | 0.70 | 0.71 | – |
| Learned, depth 1 | 12.2 | 9.1 | **1.35** | (4.77) | −19 | 63 | **1.01** | 0.82 | **2.8 (2.64)** |
| Learned, depth 2 | 11.2 | 8.6 | 1.30 | (4.37) | −18 | 64 | 0.95 | 0.83 | 1.8 (1.58) |
| Learned, depth 3 | 9.9 | 9.4 | 1.05 | (3.61) | −24 | 64 | 0.73 | 0.59 | 0.5 (0.46) |
| Multi-target, depth 2 | 8.6 | 12.2 | 0.70 | (2.44) | −44 | 66 | 0.45 | 0.61 | −0.8 (−0.37) |
| *Vehicles excluded from training and evaluation, ranked across all firms* | | | | | | | | | |
| Plain network | 14.5 | 22.3 | 0.65 | (2.24) | −68 | 37 | 0.57 | 0.31 | – |
| Learned, depth 1 | 14.1 | 22.5 | 0.63 | (2.14) | −67 | 36 | 0.55 | 0.33 | −0.4 (−0.27) |
| Learned, depth 2 | 15.7 | 22.2 | 0.71 | (2.34) | −69 | 37 | 0.63 | 0.30 | 1.2 (1.05) |
| Learned, depth 3 | 16.1 | 21.1 | 0.76 | (2.73) | −59 | 38 | 0.68 | 0.40 | 1.6 (1.14) |
| Multi-target, depth 2 | 12.7 | 26.6 | 0.48 | (1.60) | −79 | 32 | 0.42 | 0.23 | −1.7 (−0.57) |

Equal-weighted decile long–short portfolios. Plain network averaged over three training runs, graph models over two. "Ex small" excludes firms with size rank $\leq 0.2$. Last column: mean annualized difference in monthly long–short returns relative to the plain network under the same ranking rule, with Newey–West $t$-statistic. Results within 10 clusters are shown in Figure 5.

### 5.3 A learned graph helps modestly

Table 2 reports the learned-graph models. Ranked across all firms, depths 1 and 2 earn slightly more than the plain network (Sharpe ratios of 0.75 and 0.79 against 0.69) and depth 3 slightly less (0.60); no difference is significant. Ranked within 20 clusters, depth 1 earns 2.8% a year more than the plain network ($t = 2.64$), with a Sharpe ratio of 1.35 before costs and 1.01 net of an assumed 20 basis points (Figure 4b). This is the only statistically significant graph-model gain reported, and it is not robust across model configurations. Without investment vehicles, depth 3 performs best (Sharpe 0.76) and depth 1 worst (0.63), compared with 0.65 for the plain network; none of these differences is significant. The results therefore do not establish a robust predictive advantage from graph aggregation.

The return-only model's learned space is highly concentrated: averaged over January and July of 2020–2024 for operating firms, it has 1.6 effective dimensions, compared with 15.4 for raw characteristics. In the June 2024 all-firm snapshot, the corresponding values are 1.5 and 14.3 (Figure 1c), with 79% of variance in one direction. Only 10% of its neighbours persist month to

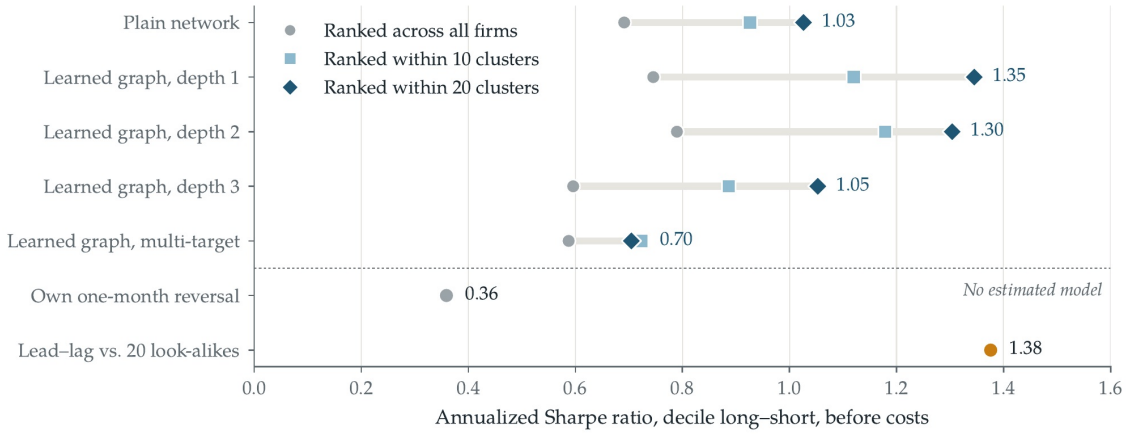



**Figure 6. Comovement with industry peers and characteristic peers.** Adjusted $R^2$ of group membership in monthly cross-sectional regressions of returns, yearly averages, operating firms, 2003–2024. (a) Raw returns. (b) Returns after removing a monthly linear fit on the 64 characteristics. Clusters are $k$-means groups with as many clusters as two-digit industries in each month. January 2003 is retained with its one-month-lagged return characteristic corrected (Section 3).

month. The multi-target model produces a higher-dimensional, more stable space (15.6 effective dimensions and 87% neighbour persistence using 12-month-averaged representations), but this does not translate into stronger residual comovement: its neighbours' residual-return correlation is 0.050, versus 0.062 for raw-characteristic neighbours and 0.075 for random same-industry peers. Its forecasting Sharpe ratio is also lower than that of the other learned-graph specifications. These diagnostics describe the geometry of the learned spaces, not evidence that the multi-target neighbours are economically more informative.

## 6 Why look-alikes add little information

If characteristic similarity captures economic links beyond the characteristics themselves, characteristic peers should exhibit stronger comovement even after controlling for those characteristics. Figure 6 tests this in monthly cross-sectional regressions over 2003–2024, using as many $k$-means clusters as there are two-digit industries in each month (about 67, excluding vehicles), so that the two groupings are equally fine.

In raw returns, cluster membership explains 7.4% of monthly cross-sectional variance, compared with 5.3% for industry; clusters dominate in 21 of 22 years (Figure 6a). A firm's return correlates 0.23 with the mean return of its ten nearest characteristic neighbours, compared with 0.13 for ten randomly selected same-industry firms. However, a linear regression on the 64 characteristics alone explains 10.9% of return variation. After removing this characteristic-based

fit, the ordering reverses: industry explains 2.1% of residual variance, compared with 0.9% for characteristic clusters (Figure 6b). In a joint regression of a firm's return on the mean returns of its industry peers and cluster peers, the loading on cluster peers falls from 0.55 to 0.10 after controlling for characteristics, while the industry-peer loading changes little, from 0.54 to 0.52. The post-control difference between the two loadings has a Newey–West $t$-statistic of $-24.3$ (six lags).

These results suggest that characteristic peers comove largely because they share characteristic exposures, consistent with Kelly et al. (2019), while industry retains incremental information beyond those characteristics. This helps explain the limited gains from graph aggregation in Section 5: when a model already observes a firm's characteristics, the representations or returns of firms with similar characteristics may add little new information. It also provides a risk-based interpretation of within-cluster ranking: comparing firms with similar characteristics reduces shared exposures and allows differences in their forecasts to determine portfolio positions.

## 7 Look-alikes as a reference group

If characteristic peers share risk exposures, a firm's return relative to those peers may isolate its idiosyncratic move and reveal overreaction. Let $\rho_{i,t}$ be firm $i$'s centred cross-sectional rank of the previous month's return, known at the start of month $t$. Define

$$LL_{i,t} = \frac{1}{k} \sum_{j \in \mathcal{N}_t(i)} \rho_{j,t} - \rho_{i,t},$$

where $\mathcal{N}_t(i)$ contains the $k$ firms most similar in characteristics. Firms that lagged their peers are bought and those that ran ahead are sold; no parameters are estimated.

With 20 neighbours and investment vehicles included, the strategy earns 12.5% annually, with 9.1% volatility, a Sharpe ratio of 1.38 ($t = 4.12$) and maximum drawdown of $-13\%$ (Table 3). Peers' returns and own reversal alone earn Sharpe ratios of 0.02 and 0.36, respectively, indicating that the signal lies in the relative comparison. Excluding investment vehicles, the Sharpe ratio is 1.12 when the neighbour pool still includes them, versus 0.85 when both the evaluated firms and neighbour pool exclude them. The strategy using 12-month-averaged multi-target neighbours

**Table 3.** Lead–lag among characteristic look-alikes, 2010–2024

| | Return (%/yr) | Vol. (%/yr) | Sharpe | ($t$) | Max DD (%) | Turn. (%/mo) | Sharpe net 20 bps† | Sharpe ex small |
|---|---|---|---|---|---|---|---|---|
| *Investment vehicles included* | | | | | | | | |
| Peers' last-month return only | 0.4 | 25.1 | 0.02 | (0.08) | −57 | 82 | −0.14 | 0.20 |
| Own reversal only | 7.2 | 20.0 | 0.36 | (1.84) | −29 | 83 | 0.16 | −0.05 |
| Lead–lag, 20 look-alikes | 12.5 | 9.1 | **1.38** | (4.12) | −13 | 88 | **0.91** | 0.78 |
| Lead–lag, multi-target, monthly | 10.6 | 7.9 | 1.33 | (5.21) | −10 | 86 | 0.81 | 0.47 |
| Lead–lag, multi-target, 12-month avg. | 12.5 | 10.5 | 1.19 | (4.10) | −11 | 86 | 0.80 | 0.57 |
| *Investment vehicles excluded* | | | | | | | | |
| Own reversal only | 6.0 | 20.2 | 0.30 | (1.45) | −32 | 83 | 0.10 | 0.09 |
| Lead–lag, 10 nbrs., pool: all firms | 9.7 | 8.7 | 1.12 | (4.22) | −14 | 89 | 0.62 | 0.76 |
| Lead–lag, 20 nbrs., pool: operating | 7.8 | 9.1 | 0.85 | (3.22) | −14 | 88 | 0.39 | 0.72 |
| Lead–lag, multi-target, monthly | 7.4 | 7.7 | 0.95 | (3.64) | −14 | 86 | 0.42 | 0.66 |
| Lead–lag, multi-target, 12-month avg. | 10.0 | 10.2 | 0.98 | (3.70) | −18 | 86 | 0.57 | 0.63 |

Equal-weighted decile long–short portfolios on the signal, before costs unless stated. †20 basis points per one-way trade is an assumed cost; no quote or order-book data were used. In the lower panel only operating firms are sorted into portfolios in every row. "Pool: all firms": each operating firm's ten look-alikes (nbrs.) are drawn from the full cross-section, including investment vehicles. "Pool: operating": the graph is built on operating firms only, so both the evaluated firms and their 20 look-alikes exclude vehicles. The two rows also differ in the number of neighbours. Multi-target: the 20 nearest firms in the learned space of the first training run, monthly or averaged over the trailing 12 months (minimum 6).

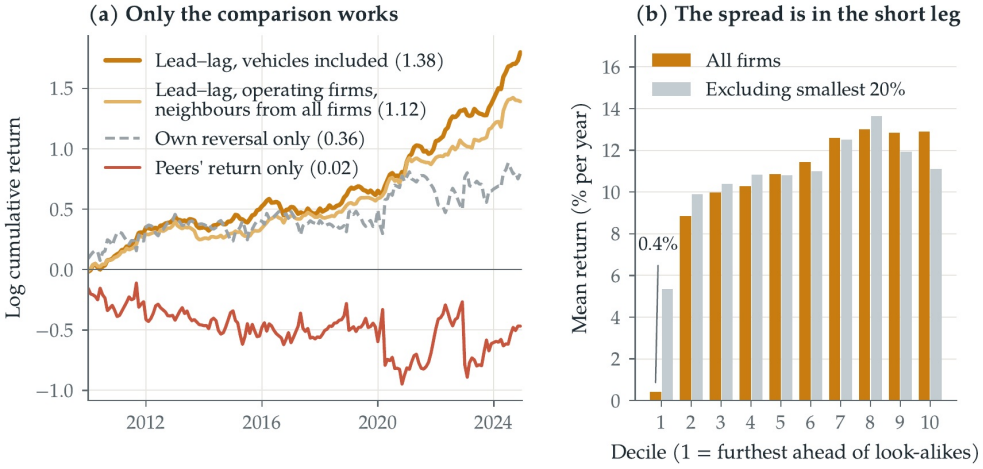



**Figure 7. Lead–lag among look-alikes,** January 2010–December 2024. (a) Log cumulative return of the lead–lag long–short portfolio and of its two components; Sharpe ratios in parentheses. The vehicles-included series sorts all firms, with 20 look-alikes drawn from all firms. The operating-firm series sorts operating firms only but draws their ten look-alikes from all firms, including investment vehicles (the 0.85 result of Table 3, which also restricts the neighbour pool, is not plotted). (b) Mean annualized return of each lead–lag decile, investment vehicles included, for all firms and excluding the bottom size quintile. Decile 1 contains the firms that ran furthest ahead of their look-alikes in the previous month.

earns 1.19, below the raw-characteristic version.

The result requires caution: the short leg earns only 0.4% annually, and monthly turnover is 88%. The net Sharpe ratio of 0.91 uses an assumed one-way transaction cost of 20 basis points; this is an illustrative adjustment, not an empirical estimate of actual trading costs. The

break-even one-way cost is 59 basis points, falling to 27 basis points when the smallest firms are excluded. A one-month execution delay reduces the Sharpe ratio to 0.26 ($t = 0.95$). Monthly data cannot distinguish genuine short-term reversal from bid–ask bounce, so the strategy's implementable profitability remains uncertain. The result is consistent with peer-relative reversal documented by Da et al. (2014) and Hameed and Mian (2015); validating its implementability would require daily data and realistic transaction-cost estimates.

## 8 Limitations and conclusion

The cluster count, learned-graph redesign and auxiliary training objectives were selected based on the 2010–2024 test period (no untouched hold-out sample) and the variability in the results of the two or three training runs per model. Factor-adjusted and value-weighted portfolios, verification of delisting-adjusted returns (Shumway, 1997), daily tests for bid–ask bounce, and transaction costs estimated from quote data remain outstanding (Fama and French, 2015; Carhart, 1997).

The findings indicate that there was no strong forecast benefit when using graph aggregation compared to a neural network with the same features. Characteristic peers are mostly a measure of exposures already observed in these characteristics, and ranking of peers within the cluster is a good way to construct a portfolio. Short-term reversal is also present in peer-relative returns, although the fact that they are hard to distinguish from a bid–ask bounce and that transaction costs are assumed implies that the economic value of the returns to implementers is uncertain. Future studies might create a graph based on data other than company information, such as product information (Hoberg and Phillips, 2016), technology (Lee et al., 2019; Kakhbod et al., 2024) and organizational capital (Chen et al., 2026) and impose strict time restriction of information protection constraints (Kakhbod and Li, 2025).

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